Google Cloud Run — Study Guide

Version 1 — baseline

Autoscaling and Concurrency

Cloud Run automatically scales the number of container instances based on incoming requests. When there are no requests, Cloud Run scales down to zero instances, eliminating costs during idle periods. You can configure a minimum number of instances to keep warm and avoid cold start latency.

Maximum instances cap how many containers Cloud Run will start, protecting downstream services. The default maximum is 100 instances per service. Each instance handles up to 80 concurrent requests by default, configurable up to 1000. Cold start latency is typically 1–3 seconds for most images.

Networking

Cloud Run services are accessible via HTTPS through an automatically provisioned URL. Custom domains can be mapped using Cloud Run domain mapping or a load balancer.

VPC Connector (Serverless VPC Access) allows Cloud Run to reach resources inside a private VPC, such as Cloud SQL or Memorystore. Ingress controls determine which traffic reaches the service: All (public internet), Internal, or Internal and Cloud Load Balancing.

Identity and Security

Cloud Run services use a service account as their identity. Best practice is to create a dedicated service account with minimal required permissions rather than using the default Compute Engine account.

Authentication can be enforced using IAM. Unauthenticated invocations must be explicitly allowed. The Cloud Run Invoker role (roles/run.invoker) grants permission to call a service.

Pricing

Cloud Run pricing is based on CPU, memory, and request count:

• CPU: charged per vCPU-second during request processing
• Memory: charged per GiB-second
• Requests: charged per million requests after free tier

The free tier includes 2 million requests, 360,000 vCPU-seconds, and 180,000 GiB-seconds per month.